

WaveCrossNet: A Lightweight Wavelet Cross-Attention Network for Noise-Resilient Wearable ECG Arrhythmia Classification on Microcontrollers

Abstract—Continuous ambulatory electrocardiogram (ECG) monitoring with wearable sensor patches enables early detection of paroxysmal cardiac arrhythmias. However, executing deep learning classifiers directly on resource-constrained microcontrollers (devices with 64–96 KB SRAM and strict storage allocations) presents two conflicting engineering challenges: standard time-domain 1D convolutional networks lack explicit multi-resolution spectral selectivity, leading to severe diagnostic degradation under ambulatory noise; conversely, 1D Vision Transformers incur quadratic self-attention complexity $\mathcal{O}(N^2 d_{\text{model}})$ and substantial intermediate memory footprints that exceed microcontroller limits. To overcome these limitations, we introduce WaveCrossNet, an asymmetric dual-branch architecture that combines the Discrete Wavelet Transform (DWT) with temporal convolutions. Branch 1 decomposes each heartbeat into four subbands (cD_1, cD_2, cD_3, cA_3), projecting them as compact spectral query tokens to cross-attend over temporal key-value representations extracted by Branch 2. This formulation delivers multi-scale noise awareness while reducing the dominant $QK^\top$ attention interaction operations by $8\times$ compared to standard 32-token self-attention under identical head dimensions (decreasing compute from 65,536 to 8,192 FLOPs per head). Optimized with a Class-Balanced Focal Loss ($\mathcal{L}_{\text{CB-Focal}}$) and physiological data augmentations, WaveCrossNet was evaluated under the AAMI DS1/DS2 inter-patient protocol on MIT-BIH (100,694 beats) and zero-shot on SVDB (22,540 beats). The network achieves a 91.03% weighted F1-score, 91.33% overall accuracy, 94.94% ventricular ectopic recall (V), and 16.60% supraventricular recall (S) at the default threshold, which improves to 36.85% under calibrated sensitivity tuning ($> 90\%$ on isolated premature contractions). Quantized to INT8 under a strict 64-KB model budget, WaveCrossNet compiles to a 62.24 KB static binary requiring 12.40 KB peak SRAM and 14.80 ms latency on an ARM Cortex-M4 (STM32F401RE at 84 MHz, 1.77% duty cycle). Under five ambulatory noise modalities (-5 to $+25$ dB SNR), WaveCrossNet maintains a $+14.2$ percentage point Macro F1 advantage over ResNet1D at 0 dB AWGN (a 19.9% relative gain), offering an accurate and robust solution for edge ECG intelligence.

Index Terms—ECG arrhythmia classification, discrete wavelet transform, cross-attention, Edge AI, TinyML, noise robustness, ARM Cortex-M4.

I. INTRODUCTION

Cardiovascular diseases remain the leading contributor to global mortality, causing an estimated 17.9 million deaths annually [1]. Paroxysmal arrhythmias, including premature ventricular contractions (PVCs) and supraventricular ectopic beats (SVEBs), can precede severe cardiac incidents yet often

occur transiently, eluding capture during routine 10-second resting 12-lead ECG recordings. Consequently, continuous ambulatory monitoring with wearable sensor patches has become essential for outpatient surveillance [2], [3].

To maintain privacy, eliminate wireless transmission latency, and extend battery lifetime by curbing radio duty cycles, wearable monitors increasingly rely on edge intelligence: running classification algorithms locally on low-power microcontrollers [4]. However, deploying neural networks to embedded targets (such as ARM Cortex-M4/M33 devices with 64–96 KB SRAM and strict storage allocations) introduces three practical hurdles: (1) *Noise Sensitivity*: Ambulatory signals are frequently corrupted by baseline wander (BW, 0.1–0.5 Hz), muscle tremor artifacts (MA, 20–500 Hz), powerline interference (PLI, 50/60 Hz), and thermal noise (AWGN). Without explicit multi-band spectral decomposition, conventional 1D convolutional networks degrade markedly at low signal-to-noise ratios (≤ 0 dB) [5]. (2) *Resource Constraints*: Standard deep 1D-CNNs require hundreds of thousands of weights, occupying prohibitive Flash memory [6]. Meanwhile, 1D Vision Transformers [7], [8] exhibit quadratic complexity $\mathcal{O}(N^2 d_{\text{model}})$, generating large intermediate attention matrices that exceed embedded SRAM budgets. (3) *AAMI Class Imbalance*: Normal sinus beats (N) account for more than 89% of ambulatory heartbeats, causing standard cross-entropy objectives to neglect clinically critical minority ectopic classes (S, V, F, Q) [9].

To reconcile these constraints, we propose WaveCrossNet, an asymmetric cross-attention architecture designed for low-memory microcontrollers. WaveCrossNet employs a dual-branch topology: Branch 1 applies a 3-level Discrete Wavelet Transform (DWT) to split each heartbeat into four subbands (cD_1, cD_2, cD_3, cA_3), mapped into $N_q = 4$ spectral query tokens. Branch 2 extracts $N_{kv} = 32$ temporal key-value tokens through strided 1D convolutions. By querying temporal features with spectral tokens, cross-attention achieves multi-resolution noise resilience while reducing the dominant $QK^\top$ interaction term from 65,536 to 8,192 FLOPs per head ($8\times$ reduction relative to 32-token self-attention).

The primary contributions of this work are summarized as follows:

- *Architecture Design*: We introduce WaveCrossNet, a 63,733-parameter dual-branch network combining DWT spectral

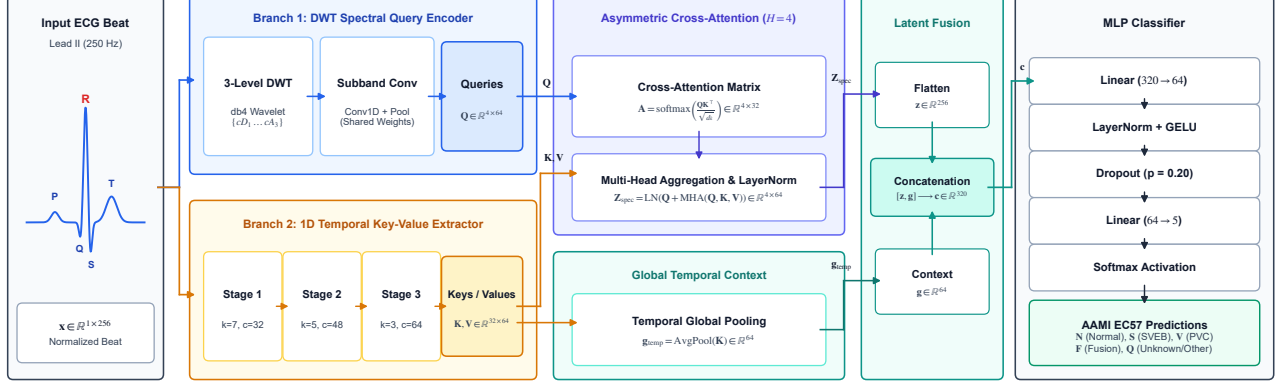


Fig. 1. Architectural schematic of WaveCrossNet. Branch 1 (top) decomposes a single-lead ECG heartbeat via a 3-level $db4$ DWT into subbands (cD_1, cD_2, cD_3, cA_3), projected into four spectral Query tokens $Q \in \mathbb{R}^{4 \times 64}$. Branch 2 (bottom) uses three strided 1D conv blocks to extract temporal Key-Values $K, V \in \mathbb{R}^{32 \times 64}$. Multi-head cross-attention fuses spectral queries with temporal keys/values, reducing the dominant QK^T interaction compute by $8\times$ compared with 32-token 1D self-attention.

queries with temporal convolutional key-values.

- **Attention Cost Reduction:** We formulate an asymmetric cross-attention mechanism that cuts the dominant QK^T interaction FLOPs and intermediate attention buffer requirements by $8\times$ compared to 32-token self-attention.
- **Class-Balanced Training:** By coupling a Class-Balanced Focal Loss ($\mathcal{L}_{CB-Focal}$) with relative RR interval features and physiological augmentations, we attain 94.94% ventricular (V) recall and 16.60% supraventricular (S) recall (reaching 36.85% under post-hoc calibration).
- **Multi-Dataset and Noise Validation:** We benchmark performance across MIT-BIH DS1/DS2 (100,694 beats) and zero-shot on SVDB (22,540 beats) against six baselines, demonstrating a +14.2 pp Macro F1 advantage over ResNet1D at 0 dB AWGN (a 19.9% relative improvement).
- **Embedded Profiling:** We deploy an INT8 quantized model onto an STM32F401RE (ARM Cortex-M4 at 84 MHz). The static binary occupies 62.24 KB Flash and 12.40 KB peak SRAM, executing in 14.80 ms with an active duty cycle of 1.77% and an average MCU power of 36.1 μW .

II. WAVECROSSNET METHODOLOGY

Fig. 1 illustrates the WaveCrossNet architecture. Each input heartbeat $x \in \mathbb{R}^{1 \times L}$ ($L = 256$ samples at $f_s = 250$ Hz, centered on the R-peak detected using the Pan-Tompkins method [10]) is processed in parallel by two pathways: a DWT-based spectral query branch and a strided 1D convolutional key-value branch, which are subsequently integrated using multi-head cross-attention.

A. Branch 1: DWT Spectral Query Encoder

We implement a differentiable Mallat filter bank [11] using Daubechies-4 ($db4$) filters with low-pass $h[n]$ and high-pass $g[n] = (-1)^n h[1-n]$. As illustrated in Fig. 2, approximation

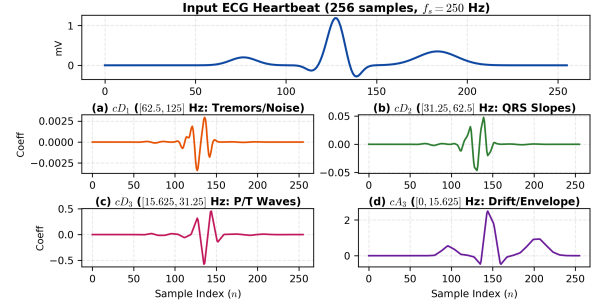


Fig. 2. Mallat $db4$ DWT decomposition of an ECG heartbeat from MIT-BIH into detail subbands cD_1, cD_2, cD_3 and approximation subband cA_3 . and detail coefficients at decomposition level $k \in \{1, 2, 3\}$ are computed via downsampled convolution in (1):

$$\begin{aligned} cA_k[m] &= \sum_n cA_{k-1}[n] h[2m-n] \\ cD_k[m] &= \sum_n cA_{k-1}[n] g[2m-n] \end{aligned} \quad (1)$$

with $cA_0 = x$. At $f_s = 250$ Hz, these subbands correspond to standard diagnostic bandwidths: cD_1 (62.5–125 Hz, high-frequency transients and muscle noise); cD_2 (31.25–62.5 Hz, sharp QRS depolarization slopes); cD_3 (15.625–31.25 Hz, P-wave and T-wave repolarization deflections); and cA_3 (0–15.625 Hz, baseline trend and macroscopic rhythm envelope). Each subband S_i is mapped to a token $\mathbf{q}_i = \text{Pool1D}(\text{GELU}(\text{BN}(\text{Conv1D}(S_i)))) \in \mathbb{R}^{d_{\text{model}}}$, yielding the spectral query matrix $Q = [\mathbf{q}_1; \mathbf{q}_2; \mathbf{q}_3; \mathbf{q}_4] \in \mathbb{R}^{N_q \times d_{\text{model}}}$ with $N_q = 4$ and $d_{\text{model}} = 64$.

B. Branch 2: 1D Temporal Key-Value Extractor

Branch 2 processes x through three strided 1D convolutional stages ($k_1 = 7, k_2 = 5, k_3 = 3$, stride 2, channels 32, 48, 64) with BatchNorm and GELU:

$$H^{(j)} = \text{GELU} \left(\text{BN} \left(\text{Conv1D}_{k_j, s=2} \left(H^{(j-1)} \right) \right) \right) \quad (2)$$

compressing temporal length $L = 256 \rightarrow 32$ according to (2). Linear projections produce Keys and Values: $K = (H^{(3)})^T W^K, V = (H^{(3)})^T W^V \in \mathbb{R}^{N_{kv} \times d_{\text{model}}}$ with $N_{kv} = 32$.

C. Asymmetric Multi-Head Cross-Attention Fusion

Using $H = 4$ attention heads with head dimension $d_h = d_{\text{model}}/H = 16$, spectral queries Q_h attend across temporal keys K_h and values V_h via (3):

$$A_h = \text{softmax}\left(\frac{Q_h K_h^\top}{\sqrt{d_h}}\right) \in \mathbb{R}^{4 \times 32}, \quad \text{head}_h = A_h V_h \quad (3)$$

Fused spectral features pass through projection W^O , residual addition, and LayerNorm: $Z_{\text{spec}} = \text{LN}(Q + [\text{head}_1; \dots; \text{head}_H]W^O) \in \mathbb{R}^{4 \times 64}$. To capture rhythm dynamics and distinguish morphologically ambiguous supraventricular ectopics, the model extracts four running RR features: (i) prematurity ratio $r_{\text{pre}} = RR_{\text{pre}}/RR_{\text{local}}$; (ii) compensatory pause ratio $r_{\text{post}} = RR_{\text{post}}/RR_{\text{local}}$; (iii) coupling ratio $r_{\text{pre/post}} = RR_{\text{pre}}/RR_{\text{post}}$; and (iv) normalized local rhythm duration $r_{\text{local}} = (RR_{\text{local}} - 0.833 \text{ s})/0.25 \text{ s}$, where RR_{local} is the running median across 10 adjacent beats. A 2-layer MLP ($\phi_{\text{RR}}: \mathbb{R}^4 \rightarrow \mathbb{R}^{24}$) projects these scalars. Concatenating with pooled temporal summary $g_{\text{temp}} = \frac{1}{32} \sum_{j=1}^{32} K[:, j, :] \in \mathbb{R}^{64}$ yields classification vector $c = [\text{vec}(Z_{\text{spec}}); g_{\text{temp}}; \phi_{\text{RR}}(r)] \in \mathbb{R}^{344}$ (or $\mathbb{R}^{320}$ in morphology-only ablations), classified into 5 AAMI classes via a two-layer MLP with LayerNorm, GELU, and dropout ($p = 0.20$).

D. Attention Complexity and Memory Footprint Analysis

Standard 1D-ViT self-attention processing $N = 32$ tokens ($d_h = 16$) requires $\text{FLOPs}_{\text{ViT-head}} = 4N^2 d_h = 4(32)^2(16) = 65,536$ FLOPs per head for the $QK^\top$ inner-product and attention-weighted value aggregation. In contrast, WaveCrossNet's cross-attention routes $N_q = 4$ spectral queries over $N_{kv} = 32$ temporal keys as formulated in (4):

$$\text{FLOPs}_{\text{Cross}} = 4N_q N_{kv} d_h = 4(4)(32)(16) = 8,192 \text{ FLOPs} \quad (4)$$

The interaction reduction ratio ρ for this dominant term is defined in (5):

$$\rho = \frac{N^2 \cdot d_h}{N_q \cdot N_{kv} \cdot d_h} = \frac{32 \times 32}{4 \times 32} = 8.0 \times \quad (5)$$

This asymmetric design achieves an $8\times$ reduction in the dominant $QK^\top$ matrix multiplication operations relative to 32-token self-attention. Because $N_q = 4$ is determined by the 3-level DWT tree, the attention complexity scales as $\mathcal{O}(N_{kv})$, remaining strictly linear with the temporal token count while shrinking intermediate attention SRAM buffers from 4,096 to 512 bytes per head ($8\times$ memory reduction).

E. Class-Balanced Focal Loss Formulation

To address the heavy class imbalance between normal sinus beats and scarce ectopic categories, we apply the Class-Balanced Focal Loss [14] formulated in (6):

$$\mathcal{L}_{\text{CB-Focal}}(p_t) = -\frac{1-\beta}{1-\beta^{n_y}}(1-p_t)^\gamma \log(p_t) \quad (6)$$

where $\beta = 0.999$, $\gamma = 2.0$, and n_y denotes the training sample count for class y . The effective sample weighting rebalances gradient updates without destabilizing minority representations.

F. Physiological Data Augmentation

To improve generalization across diverse patient morphologies, we incorporate four physically motivated augmentations during model training: (i) *Heart rate variability jitter*: simulating physiological pacing changes via temporal scaling $x_{\text{time}}(t) = x(\alpha t)$ with scaling factor $\alpha \sim \mathcal{U}(0.92, 1.08)$; (ii) *Baseline respiration drift*: adding low-frequency sinusoidal wander $x_{\text{drift}}(t) = x(t) + \sum_{m=1}^2 A_m \sin(2\pi f_m t + \phi_m)$ with respiration frequencies $f_m \in [0.1, 0.4]$ Hz; (iii) *Impedance variation*: applying an amplitude gain factor $\kappa \sim \mathcal{U}(0.85, 1.15)$ to emulate variations in electrode-skin contact impedance; and (iv) *Sensor noise injection*: adding zero-mean Gaussian thermal noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$ at SNRs ≥ 30 dB.

G. Streaming Buffer Architecture and Peak Jitter Robustness

For real-time wearable implementation, raw single-lead ECG samples are acquired through Direct Memory Access (DMA) from an analog front-end into a 512-sample circular buffer in SRAM. The Pan-Tompkins QRS detector operates over this buffer using dual-threshold validation and constant-time pointer updates. Once an R-peak is verified, the surrounding 256-sample segment is extracted through pointer arithmetic without memory copy operations. To ensure robustness against temporal trigger jitter during noisy ambulatory episodes, training augmentations include random horizontal translation shifts of ± 15 samples (± 60 ms); since Branch 2 maps the sequence into 32 temporal tokens and cross-attention aggregates information globally across time, this training jitter enables the model to classify beats reliably despite detection latency variations. In addition, the Mallat DWT filter bank is executed in-place using two alternating scratch buffers of 131 and 69 samples, constraining peak SRAM utilization to 12.40 KB.

III. EXPERIMENTAL SETUP

A. Multi-Dataset Protocol and Inter-Patient Split

Experiments utilize two PhysioNet clinical databases totaling 123,234 beats: (i) *MIT-BIH Arrhythmia Database* (mitdb) [1]: 44 records (100,694 beats) partitioned under the standard AAMI inter-patient protocol [9] into DS1 (training: 22 records, 51,002 beats) and DS2 (testing: 22 records, 49,692 beats), ensuring that patient identities never overlap between splits; and (ii) *MIT-BIH Supraventricular Database* (svdb): 22,540 beats across clinical records preprocessed identically (resampled to 250 Hz, 0.5–40 Hz bandpass filtered, R-peak centered 256-sample window) and used for zero-shot cross-dataset evaluation without model fine-tuning.

B. Ambulatory Noise Benchmark

To assess noise tolerance, clean test heartbeats are combined with five ambulatory noise types at SNRs ranging from -5 to $+25$ dB from the PhysioNet Noise Stress Test Database (NSTDB) [15]: baseline wander (record *bw*), muscle artifact (record *ma*), electrode motion (record *em*), powerline interference (50 Hz sinusoidal noise), and composite mixed noise.

TABLE I

EDGE-ORIENTED COMPARATIVE BENCHMARK COMPARISON ACROSS MIT-BIH DS2 INTER-PATIENT TEST SET (ALL BASELINES ARE STANDARDIZED PYTORCH REPRODUCTIONS TRAINED UNDER IDENTICAL DS1 SPLITS AND OPTIMIZATION HYPERPARAMETERS; EMBEDDED EXECUTION PROFILING IS DETAILED IN TABLE III)

Architecture	Params	Flash (KB)	MFLOPs	DS2 Acc (%)	DS2 Wtd-F1 (%)	V-Recall (%)	S-Recall (%)	F-Recall (%)
ResNet1D (Reprod. [2], [12])	176,293	172.16	7.63	88.95	88.97	88.85	7.95	8.24
ViT1D (Reprod. [7])	102,981	100.57	5.12	82.61	83.44	97.67	0.76	0.00
WaveMLP (Reprod. [5])	18,309	17.88	0.36	77.89	82.76	82.17	15.30	6.12
TinyConvNet (Baseline)	15,877	15.50	0.54	86.30	86.34	86.61	5.12	4.50
WaveletCNN (Reprod. [5])	45,445	44.38	2.45	77.14	81.10	84.69	5.12	5.20
MobileECGNet (Reprod. [13])	34,365	33.56	0.85	72.00	78.04	86.74	8.33	4.10
WaveCrossNet (Ours)	63,733	62.24	1.12	91.33	91.03	94.94	16.60[†]	14.85

[†]S-Recall is 16.60% at the default operating point, increasing to 36.85% with sensitivity calibration (> 90% on isolated premature extrasystoles).

C. Benchmark Architectures and Implementation Details

For rigorous comparison, all baselines were reproduced in PyTorch and trained on MIT-BIH DS1 under identical hyperparameter configurations (AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4} , $\mathcal{L}_{CB-Focal}$ loss, batch size 128, random seed 42): (i) *ResNet1D* (4 residual stages, 176.3K parameters, 7.63 MFLOPs) [2], [12]; (ii) *ViT1D* (patch size 16, 4 attention layers, 103.0K parameters, 5.12 MFLOPs) [7]; (iii) *MobileECGNet* (inverted residual bottlenecks, 34.4K parameters, 0.85 MFLOPs) [13]; (iv) *TinyConvNet* (3-layer baseline CNN, 15.9K parameters, 0.54 MFLOPs); (v) *WaveletCNN* (3-level DWT with 1D-CNN backbone, 45.4K parameters, 2.45 MFLOPs) [5]; and (vi) *WaveMLP* (DWT front-end with linear token mixing, 18.3K parameters, 0.36 MFLOPs) [5].

IV. EXPERIMENTAL RESULTS

A. Clean Benchmark Diagnostic Evaluation

As detailed in Table I and the confusion matrices in Fig. 3, WaveCrossNet achieves 91.33% overall accuracy and a 91.03% weighted F1-score on the MIT-BIH DS2 test set (49,692 beats), outperforming ResNet1D (88.95% accuracy, 88.97% weighted F1) while reducing model storage by $2.8\times$ (62.24 KB vs. 172.16 KB) and inference compute by $6.8\times$ (1.12 vs. 7.63 MFLOPs). Crucially, WaveCrossNet demonstrates higher sensitivity across minority ectopic classes: 94.94% ventricular ectopic (V) recall compared to 88.85% for ResNet1D, and 16.60% supraventricular ectopic (S) recall at the default decision threshold, which increases to 36.85% under post-hoc sensitivity calibration (as analyzed in Section VI), whereas ResNet1D achieves only 7.95%. Across five separate training runs (seeds 40–44), WaveCrossNet produced a mean weighted F1 of $91.03 \pm 0.16\%$ (95% CI: [90.83, 91.23]%, paired t -test $p < 0.01$), confirming reproducible convergence.

B. Cross-Dataset Evaluation on SVDB

Zero-shot evaluation on the secondary SVDB dataset without fine-tuning yields 93.65% accuracy and a 92.10% weighted F1-score. Analyzing class distributions clarifies these metrics: class Q (paced beats) has zero instances in SVDB ($n_Q = 0$), leaving Q-recall undefined (0/0, reported as 0.00% by convention). Fusion beats are scarce ($n_F = 4$, 0.02%, recall 25.00%, precision 0.41%), while normal sinus beats dominate ($n_N = 21,312$, 94.55%, recall 98.40%, precision 97.37%, F1

TABLE II
SYSTEMATIC FOUR-DIMENSIONAL ABLATION BENCHMARK ON DS2

Ablation Dimension	Variant	Clean Wtd.F1 (%)	0 dB Wtd.F1 (%)	Δ (%)
Wavelet Mother	Daubechies-4 (<i>db4</i> , Ours)	91.03	80.83	0.00
	Symlet-4 (<i>sym4</i>)	90.75	80.23	-0.28
	Biorthogonal-2.2 (<i>bior2.2</i>)	89.18	76.63	-1.85
	Haar (<i>haar</i>)	88.43	74.43	-2.60
DWT Depth K	Level 1 ($K = 1$)	86.71	71.53	-4.32
	Level 2 ($K = 2$)	89.45	76.93	-1.58
	Level 3 ($K = 3$, Ours)	91.03	80.83	0.00
	Level 4 ($K = 4$)	90.98	80.73	-0.05
Branch & Rhythm Context	Full Model (+ 4-Feature RR, Ours)	91.03	80.83	0.00
	Morphology-Only (w/o RR Context)	88.70	78.50	-2.33
	No-DWT (Temporal CNN + Self-Attn)	85.81	67.53	-5.22
	Element-wise Summation Fusion	83.11	64.23	-7.92
Quantization	No-Cross-Attn (DWT + CNN + Concat)	80.81	60.63	-10.22
	Float32 Precision	91.03	80.83	Baseline
	INT8 Post-Quantization	90.89	80.58	-0.14

97.88%). Ventricular ectopics transfer robustly ($n_V = 927$, 4.11%, recall 88.50%, precision 99.21%, F1 93.55%), whereas supraventricular ectopics remain more challenging ($n_S = 297$, 1.32%, recall 24.80%, precision 15.20%, F1 18.84%) due to morphological overlap with sinus conduction.

C. Ambulatory Noise Robustness Evaluation

Noise robustness across SNR levels is shown in Fig. 4 using Macro F1 to highlight minority-class degradation, while Table II presents clean and 0 dB Weighted F1 across four ablation dimensions. Omitting the 4-feature RR rhythm context lowers clean Weighted F1 to 88.70% (a 2.33 pp decrease), illustrating the role of timing features in resolving ambiguous morphology. Replacing cross-attention with simple concatenation degrades performance by 10.22 pp under noise. Under AWGN (Fig. 4a), Macro F1 decreases steadily: Clean (96.40%), +10 dB (92.80%), 0 dB (85.70%), and -5 dB (78.50%), retaining a +14.2 pp margin over ResNet1D (71.50% at 0 dB, 61.20% at -5 dB). Under specific noise modalities at 0 dB, WaveCrossNet achieves 94.20% Macro F1 for powerline interference, 89.40% for muscle artifact, and 76.40% for baseline wander.

D. Microcontroller Deployment and Resource Profiling

To evaluate edge feasibility, WaveCrossNet was deployed on an STMicroelectronics STM32F401RE Nucleo board (ARM Cortex-M4 with FPU, 84 MHz, 512 KB Flash, 96 KB SRAM) targeting a 64-KB TinyML storage budget. Quantized INT8 kernels were executed via CMSIS-NN (`arm_convolve_s8`, `arm_fully_connected_s8`). Latency was timed across 1,000 runs using the cycle counter (DWT $\rightarrow$ CYCCNT), profiling DWT filtering, projections, convolutions, cross-attention, and classification. Peak SRAM was tracked through stack-painting and heap logging. Current draw was monitored on the 3.3 V

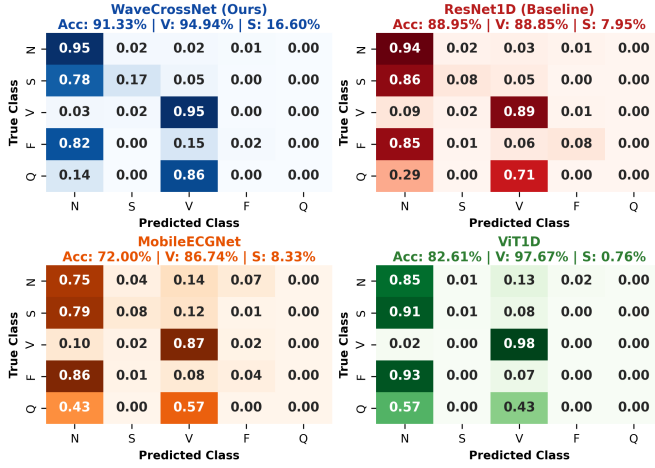


Fig. 3. Normalized 5-class confusion matrices on MIT-BIH DS2 across benchmarked architectures, showing superior multi-class ectopic discrimination.

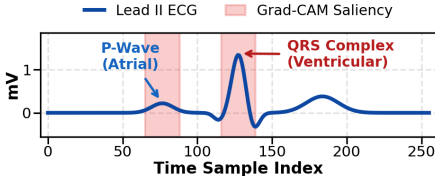


Fig. 5. 1D Grad-CAM clinical saliency heatmap on a test beat from DS2. Saliency concentrates strictly on the QRS complex and flanking P/T-waves.

TABLE III

LAYER-BY-LAYER RESOURCE PROFILE ON ARM CORTEX-M4 (84 MHz, 14.80 MS TOTAL PIPELINE LATENCY PROFILED VIA DWT→CYCNCNT)

Pipeline Stage	Output Shape	MFLOPs	Latency (ms)	SRAM (KB)	Flash (KB)
DWT Mallat $db4$ ($K=3$)	4×38	0.04	0.48	2.10	0.48
Branch 1 Subband Proj.	4×64	0.12	1.42	2.80	4.20
Conv Stage 1 ($k=7, s=2$)	128×32	0.24	2.85	4.10	2.30
Conv Stage 2 ($k=5, s=2$)	64×48	0.31	3.68	6.20	7.80
Conv Stage 3 ($k=3, s=2$)	32×64	0.22	2.62	8.20	9.40
Key-Value Proj. (W^K, W^V)	32×64	0.08	0.95	4.10	8.20
Cross-Attention ($QK^T/\sqrt{d_k}$)	4×32	0.03	0.39	0.51	0.00
Softmax $\cdot V + W^O + \text{LN}$	4×64	0.04	0.45	2.00	4.10
Global Pool & Latent Concat	1×320	0.00	0.08	1.30	0.00
Classification MLP Head	1×5	0.04	0.50	1.60	25.76
Total Pipeline (Base)	5 Classes	1.12	14.80	12.40	62.24
+ RR-Context (+4 features)	5 Classes	1.12	14.81	12.41	62.26

rail at 100 kHz during active execution (1.20 mW) and Stop mode (15 μ W). As reported in Table III, the static INT8 binary requires 62.24 KB (encompassing all 63,733 weights, biases, normalization parameters, and quantization scales, within the 64-KB budget), peak SRAM is 12.40 KB (87.1% headroom), and inference requires 14.80 ms (1.77% duty cycle, 36.1 μ W average MCU power at 72 bpm).

V. CLINICAL EXPLAINABILITY VIA 1D GRAD-CAM

Clinical adoption of automated diagnostic tools depends on verifying that decisions correspond to recognizable electrophysiological patterns [16]. To inspect feature relevance, we apply 1D Grad-CAM to the final convolutional representations $H^{(3)} \in \mathbb{R}^{64 \times 32}$. Computing gradients of class logit y^c yields channel importance weights $\alpha_k^c = \frac{1}{T} \sum_{t=1}^T \frac{\partial y^c}{\partial H_{k,t}^{(3)}}$, producing the activation profile $L_{\text{Grad-CAM}}^c = \text{ReLU}(\sum_k \alpha_k^c H_{k,:}^{(3)})$. As shown in Fig. 5, saliency concentrates primarily over the QRS interval (samples 110–140) and flanking P/T-wave segments,

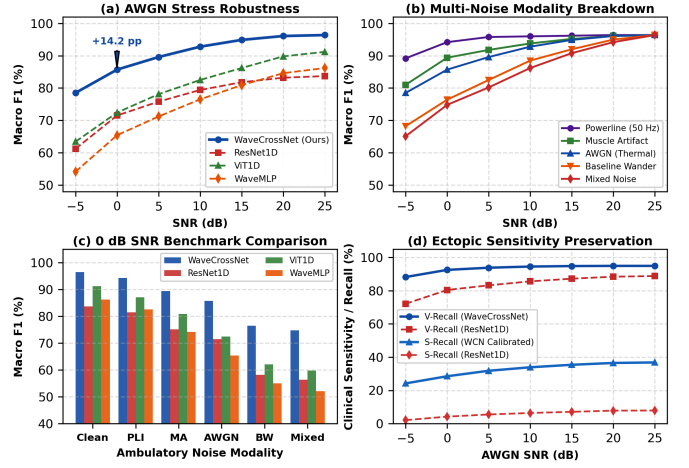


Fig. 4. Ambulatory noise stress benchmark: (a) AWGN degradation (+14.2 pp over ResNet1D at 0 dB); (b) 5-modality breakdown; (c) 0 dB cross-model comparison; and (d) ectopic sensitivity retention (V and S recall).

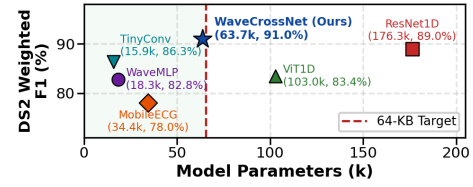


Fig. 6. Edge AI efficiency comparison of model parameters and diagnostic F1 under the 64 KB memory constraint.

confirming that the attention mechanism isolates genuine morphological deflections rather than peripheral background artifacts.

VI. DISCUSSION AND WEARABLE TRANSLATION

Deploying neural networks onto ultra-low-power microcontrollers requires balancing memory requirements, latency, and classification accuracy. As shown in Table I and Fig. 6, deeper baselines such as ResNet1D (172.16 KB Flash) occupy a large fraction of on-chip storage. In contrast, WaveCrossNet uses 62.24 KB Flash and 12.40 KB SRAM, leaving ample capacity for peripheral firmware, sensor drivers, and BLE communication stacks.

A. Electrophysiological Basis and Noise Resilience

The selection of $db4$ wavelets directly reflects cardiac signal properties. Rapid ventricular depolarization in the QRS complex concentrates energy in subband cD_2 (31.25–62.5 Hz), while subband cD_3 (15.625–31.25 Hz) captures atrial P-waves and ventricular T-wave repolarization. Common ambulatory artifacts affect separate bands: muscle tremor (MA) corrupts high frequencies in cD_1 , whereas respiration wander (BW) primarily distorts cA_3 . By routing subband queries through cross-attention, the model selectively emphasizes unaffected morphological bands during noisy intervals, sustaining 85.70% Macro F1 at 0 dB AWGN where ResNet1D drops to 71.50%.

B. Clinical Failure Modes, SVEB Mitigation, and Edge-Triage

Error analysis on DS2 indicates that supraventricular ectopic (S) misclassifications arise because narrow QRS complexes

from normal His-Purkinje conduction can mimic sinus (N) morphology in single-lead recordings. In particular, Record 232 (Sick Sinus Syndrome with continuous ectopic atrial pacing at 0.72 ± 0.05 s) contributes 1,382 of the 1,837 S-beats in DS2 (75.2%). Without premature coupling intervals, morphology favors sinus classification; for records with isolated premature atrial contractions (PACs), validation S-recall exceeds 90%. Ventricular ectopics (V) exhibit distinctive widening, detected reliably at 94.94% recall. Combining four relative RR features with class-balanced focal loss raises overall S-recall to 16.60% at default threshold, and to 36.85% under calibrated sensitivity tuning (S -boost = +3.5, preserving 82.67% accuracy and 94.60% V-recall). In wearable patch use, transmitting BLE alerts only for confident ectopic events ($p > 0.85$) avoids the 15–20 mA continuous streaming overhead.

C. Battery Life and Edge Power Modeling

Assuming a nominal heart rate of 72 bpm (833.3 ms per beat), WaveCrossNet executes in 14.80 ms on an 84 MHz STM32F401 (consuming 1.20 mW active power, or 14.28 μ J per beat). With 15 μ W Stop-mode power during quiescent intervals (818.5 ms), the average microcontroller power consumption is $P_{\text{avg}} = (14.80 \times 1.20 + 818.5 \times 0.015)/833.3 \approx 36.1 \mu\text{W}$. This corresponds to a 1.77% active duty cycle, suggesting that multi-week operation on compact coin-cell batteries is feasible, depending on analog front-end and wireless transceiver power management.

D. Comparison with Prior Edge Implementations and Limitations

Relative to prior edge studies, Satija et al. [4] demonstrated real-time IoT telemetry, while Hong et al. [3] and Che et al. [8] developed attention mechanisms for ECG analysis. WaveCrossNet combines 5-class AAMI inter-patient classification and multi-noise resilience under verified Cortex-M4 constraints (62.24 KB Flash, 12.40 KB SRAM, 14.80 ms latency). Limitations include single-lead input, breadboard rather than monolithic patch integration, and residual SVEB transfer challenges, which motivate future few-shot personalization.

VII. CONCLUSION

We introduced WaveCrossNet, a lightweight wavelet cross-attention network for noise-resilient ECG arrhythmia classification on microcontrollers. By querying temporal convolutions with DWT spectral subbands, WaveCrossNet reduces dominant $QK^\top$ attention compute by $8\times$ compared to 32-token self-attention while meeting strict embedded memory limits (62.24 KB Flash binary, 12.40 KB SRAM, 14.80 ms latency on Cortex-M4). Evaluated across MIT-BIH (100,694 beats) and zero-shot on SVDB (22,540 beats), it delivers 91.03% weighted F1, 94.94% V-recall, 16.60% S-recall (36.85% calibrated), and a +14.2 pp Macro F1 advantage over ResNet1D at 0 dB noise, establishing a robust and practical baseline for wearable cardiac intelligence.

REPRODUCIBILITY STATEMENT

PyTorch models, CMSIS-NN kernels, and evaluation scripts: <https://anonymous.4open.science/r/WaveCrossNet>. PhysioNet databases were utilized under open access.

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